INTRODUCTION

We have already learnt the general characteristics of p-block elements and the first two group namely icosagens (boron group) and tetragens (carbon group) in the previous unit. In this unit we learn the remaining p-block groups, pnictogens, chalcogens, halogens and inert gases.

3.1 Group 15 (Nitrogen group) elements:

3.1.1 Occurrence:

About 78 % of earth atmosphere contains dinitrogen (N_2) gas. It is also present in earth crust as sodium nitrate (Chile saltpetre) and potassium nitrate (Indian saltpetre). The 11th most abundant element phosphorus, exists as phosphate (fluroapatite, chloroapatite and hydroxyapatite). The other elements arsenic, antimony and bismuth are present as sulphides and are not very abundant.

3.1.2 Physical properties:

Some of the physical properties of the group 15 elements are listed below

Table 3.1 Physical properties of group 15 elements

Property	Nitrogen	Phosphorus	Arsenic	Antimony	Bismuth
Physical state at 293 K	Gas	Solid	Solid	Solid	Solid
Atomic Number	7	15	33	51	83
Isotopes	^{14}N, ^{15}N	^{31}P	^{75}As	^{121}Sb	^{209}Bi
Atomic Mass ($g.mol^{-1}$ at 293 K)	14	30.97	74.92	121.76	209.98
Electronic configuration	$[He]2s^2\ 2p^3$	$[Ne]3s^2\ 3p^3$	$[Ar]3d^{10}\ 4s^2\ 4p^3$	$[Kr]4d^{10}\ 5s^2\ 5p^3$	$[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^3$
Atomic radius (Å)	1.55	1.80	1.85	2.06	2.07
Density ($g.cm^{-3}$ at 293 K)	1.14×10^{-3}	1.82 (white phosphorus)	5.75	6.68	9.79
Melting point (K)	63	317	Sublimes at 889	904	544
Boiling point (K)	77	554		1860	1837

3.1.3 Nitrogen:

Preparation:

Nitrogen, the principal gas of atmosphere (78 % by volume) is separated industrially from liquid air by fractional distillation

Pure nitrogen gas can be obtained by the thermal decomposition of sodium azide about 575 K

$$2NaN_3 \longrightarrow 2\ Na + 3N_2$$

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